
Managing intellectual property in R&D alliances

Bing-Sheng Teng

Department of Strategic Management and Public Policy
School of Business, George Washington University
2115 G Street, NW, Washington, DC 20052, USA
Fax: (202) 994-8113
E-mail: teng@gwu.edu

Abstract: This paper deals with the management of Intellectual Property (IP) in R&D alliances, with a focus on two key IP types – patents and trade secrets. Since IP is both inputs to and outputs of R&D alliances, the key is to simultaneously achieve two objectives – IP protection and IP generation. As research inputs, IP could be devalued by partners' opportunistic learning. Thus, IP needs to be carefully selected and protected through mechanisms such as equity arrangement, non-disclosure and non-compete agreements, and monitoring and auditing.

Meanwhile, the performance of R&D alliances is mostly measured by their IP outputs. Partner firms face new issues, such as IP classification and ownership. They must decide whether to patent or keep their R&D outputs confidential. In making IP ownership decisions, partners need to consider R&D outputs' potential as the firm's core competencies and the risk of patent interference.

Keywords: Intellectual Property (IP); R&D alliance; opportunistic learning; patents; trade secrets.

Reference to this paper should be made as follows: Teng, B-S. (2007) 'Managing intellectual property in R&D alliances', *Int. J. Technology Management*, Vol. 38, Nos. 1/2, pp.160–177.

Biographical notes: Dr. Bing-Sheng Teng is Associate Professor of Strategic Management and Public Policy at the School of Business of George Washington University. He received his PhD in Strategic Management from City University of New York in 1998. Dr. Teng's current research interests include strategic alliances, corporate entrepreneurship and international management. His work has appeared in top journals, such as the *Academy of Management Review*, *Organisation Science*, *Organisation Studies*, *Journal of Management*, and *Academy of Management Executive*. In 2003, he was named Crain Research Scholar at the School of Business, George Washington University.

1 Introduction

For years, British phonemaker Sendo worked with Microsoft in an R&D alliance for developing web-enabled cell phones that run Microsoft software (Buckman, 2002). Now the two are in court for an Intellectual Property (IP) misappropriation suit against Microsoft. According to Sendo, after the delivery of its phones was delayed in August 2001 owing to bugs in Microsoft's software, Microsoft requested a full review of Sendo's phone project and thus had unlimited access to Sendo's confidential technical information.

It was alleged that Microsoft then dumped Sendo and used the ill-gotten trade secrets to launch its own cell phones in October 2002 with a new partner, High Tech Computer of Taiwan. Furthermore, Sendo charged that Microsoft unveiled its phone ahead of Sendo in a deliberate effort to drive Sendo into bankruptcy. The terms of the alliance agreement were such that, in case of Sendo's becoming bankrupt, Microsoft would get a royalty-free licence to use Sendo's IP related to its phones.

This case illustrates the complexity of managing IP in R&D alliances, such as choosing a trustworthy partner, exposing the right amount of IP, deterring opportunistic learning, maintaining sufficient IP control and constructing agreements that effectively allocate IP rights. This paper intends to shed some light on these interrelated issues.

2 R&D alliances and IP management

Technological innovation has been the driving force for evolution in many industries. The success or failure of a firm's R&D activities often has serious implications for its overall performance. As such, firms have aggressively developed a variety of ways to improve their R&D performance and productivity. One of the most successful and widely employed approaches has been a cooperative strategy – R&D alliances (Inkpen, 1998).

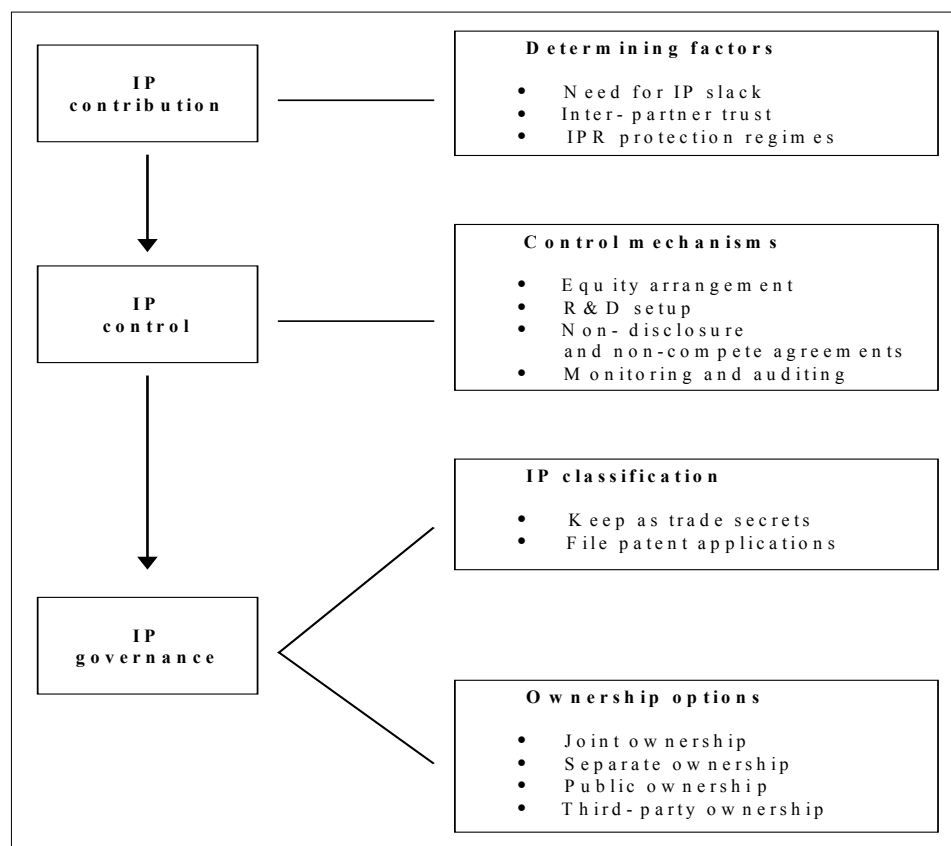
R&D alliances broadly refer to interfirm collaborations in conducting R&D. Based on their alliance agreements, partner firms contribute certain R&D-related resources in order to engage in joint R&D. In doing so, they hope to overcome their individual limitations in R&D capabilities and enhance their R&D success rate. In addition, partner firms could jointly endorse R&D outputs, thereby creating a network externality effect that increases the innovations' chance for market acceptance. Network externality refers to a situation in which the value of a good (such as fax machines) increases when there are more users.

R&D alliances have been formed in many R&D-intensive industries, including biotech-pharmaceutical, computer, semiconductor, electronics and software. While a few alliances were successful, the cooperative approach often has a high failure rate, due to difficulties such as shared control and interpartner competition (Park and Russo, 1996). As the Sendo-Microsoft case shows, a challenging aspect of such alliances is the management of IP. Indeed, R&D alliances are all about IP – IP is both key inputs to and outputs of R&D alliances. While the purpose is to use existing IP to generate additional IP, the process can be problematic since alliances represent incomplete contracting between firms (Chi and Roehl, 1997). The ownership of IP can be blurred in the process and may present a significant challenge to firms.

Given the challenge, authors have written on the topic of IP protection in strategic alliances. Nevertheless, there are three notable gaps in this literature. First, the context is often strategic alliances in general rather than R&D alliances (Baughn *et al.*, 1997; Norman, 2001). Strategic alliances include all types of interfirm collaboration, with or without an R&D component. As a subset of strategic alliances, R&D alliances have unique characteristics regarding IP management (*i.e.*, being both key inputs and outputs) that deserve separate consideration. In addition, existing papers have mostly focused on the protection rather than the management of IP. Some experts even suggest that the purpose is to “hold back as much IP and still run the operation” (Kaltenheuser, 1999, p.54). While IP protection is important, it is but one aspect of IP management in R&D alliances. Since IP is both R&D inputs and outputs, it is as important to leverage one’s existing IP as to protect it (Williams and Bukowitz, 2001). Thus, the unique challenge in R&D alliances is to balance IP protection and generation in one setting. These two objectives must be achieved simultaneously without significantly compromising either one. This is a new issue for most firms, who tend to emphasise IP protection in alliances.

Another new issue (and a gap in the literature) is the use of trade secrets in alliances. Since trade secrets have a relatively short history of legal status in the USA, many firms are not familiar with their implications, particularly in the context of alliances. The conventional wisdom seems to be avoiding the use of trade secrets in alliances owing to opportunistic learning. In contrast, this paper will show that trade secrets could be preferred in specific alliance situations.

Figure 1 A three-step framework of IP management in R&D alliances



The purpose of this paper is to broadly examine various aspects of IP management in R&D alliances. As shown in Figure 1, IP management can be viewed as a three-step process, namely, IP contribution, IP control and IP governance. The next section starts with a discussion on the various types of IP, with a focus on patents and trade secrets.

3 Types of IP

Generally, IP has four forms – trademarks, copyrights, patents and trade secrets. While trademarks are commercial identities of firms, copyrights are concerned with literary or artistic expressions. For R&D alliances, patents and trade secrets are more relevant and they have distinctive features. Firms obtain patents through applications with the US Patent and Trademark Office (USPTO). Once a technology or design is patented, relevant information regarding the patent will be disclosed and become public knowledge. The patent holder will have exclusive rights to the use of the technology defined in the patent for a certain period, which in the USA is currently 20 years from the date of patent filing.

The rationale for granting patents is to provide incentives for innovation. Without exclusive rights, innovators will be concerned about imitations that undermine the commercial benefits of innovations. As such, society will suffer from a lower level of R&D efforts. In the USA, the past two decades have seen a significant increase in patent applications and number of patents granted, partly due to institutional changes in the US patent system (Jaffe and Lerner, 2004). First, from 1982, all patent cases are heard in a single federal appeals court, rather than the twelve regional courts. Second, in 1992, the US Congress changed the fee structure of USPTO so that it would operate on the fees paid by patent applicants. These changes are believed to tilt the system to favour patent holders, *i.e.*, making it easier to obtain patents and to sue others for patent infringement.

In contrast to patents, there is no application process for trade secrets. Owners of the proprietary information simply hold on to the information and take steps to protect it from unintended disclosure (Hemphill, 2004). While trade secrets were first asserted in the English common law in the sixteenth century, in the USA they were legally recognised in 1996 when Congress passed the Economic Espionage Act (EEA). Consequently, most states have enacted the Uniform Trade Secret Act (UTSA) to protect trade secrets. According to UTSA, trade secrets are:

“...information including a formula, pattern, compilation, program, device, method, technique or process that derives independent economic value, actual or potential, from not being generally known to, and not being readily ascertainable by proper means by, other persons ...”

Based on the UTSA definition, trade secrets have three important characteristics:

- 1 a compilation of business information
- 2 economic value from its secrecy
- 3 efforts to maintain its secrecy.

Not only are they business information, but also they must have potential or real economic value when not generally known to others. In addition, reasonable steps must be taken to guard against the disclosure of the information so that others cannot obtain it

through legal means. Over the years, the losses of trade secrets have been staggering. According to one estimate, between January 1997 and June 1998 Fortune 1000 firms lost trade secrets worth \$45 billion (Jesitus, 2000).

This paper will focus on patents and trade secrets, as these are most relevant in R&D alliances. The two differ significantly in terms of legal protection. Since patents are exclusive, others cannot use the technology contained in a patent before its expiration. In contrast, trade secrets may be independently developed and then used by others. For example, the US Supreme Court ruled that it is legal to reverse engineer a publicly sold, non-patented product (Fraumann and Koletar, 1999). Thus, there is no privilege of exclusivity in trade secrets. While patents offer better protection, their drawback is that much relevant information has to be disclosed in the patenting process. As such, this process may give away valuable information to competitors and others. In recent years, the reliance on trade secrets rather than patents has been on the rise (Carayannis and Alexander, 1999). Based on a large-scale survey of US R&D labs, trade secrets are believed to be more important than patents as a mechanism to protect innovations (Cohen *et al.*, 2000).

4 IP contribution: what to share

In forming R&D alliances, one of the first and most important IP decisions is what to share with partners. R&D alliances call for partner firms to contribute cutting-edge technologies and knowledge. Before making the selection decision, partner firms need to do their homework regarding their IP. They should take stock of their existing IP and collect all agreements involving them. These measures help firms avoid disputes regarding IP ownership in alliances. In addition, firms should also evaluate the importance of their IP, through approaches such as the VRIO framework of the resource-based view (Barney, 1991). According to this approach, a particular resource (such as an IP) is more likely to be a core competency if it is valuable, rare, less imitable and with organisational support. The goal is to develop a good understanding of both partners' IP profiles before forming R&D alliances. These are necessary steps since a firm may lose its most valuable IP through opportunistic learning or unauthorised knowledge transfer between partner firms (Larsson *et al.*, 1998).

4.1 Opportunistic learning

Interfirm learning is a key objective and aspect of many strategic alliances, including R&D alliances (Doz, 1996). Partners not only learn from the collaboration (in terms of technology, know-how, *etc.*), they also learn how to collaborate in alliances (Gulati, 1999; Powell, 1998). Learning offers a convenient and effective way to obtain valuable knowledge otherwise difficult to develop in-house. Alliance partners use interfirm learning to enhance their resource profiles and ability to compete in the marketplace. Learning could be a double-edged sword for alliance partners since it can both improve and undermine their competitiveness. While learning from others is usually beneficial, being a source for learning may result in losing one's core competencies.

Partner firms attempt to regulate interfirm learning by specifying appropriate knowledge transfer in alliances. Learning in alliances can thus be classified as legitimate learning and opportunistic learning. While legitimate learning is open and authorised, opportunistic learning is secretive and without the consent of the knowledge source. Opportunistic learning is an important hidden agenda in alliances (Hamel, 1991), since it could be the only way to obtain the most sensitive and valuable type of information. Opportunistic learning is particularly common in R&D alliances, in which technology and knowledge exposure is high.

Opportunistic learning may take place even in the process of forming R&D alliances. Consider the experience of Flotec, Inc., a company that makes oxygen regulators for medical use (Halligan, 2001). In 1995 Flotec negotiated with Southern Research, Inc. (SRI) on the joint development and production of certain regulator components. After getting detailed technical information from Flotec, SRI made a proposal that was deemed too costly by Flotec. After the negotiation was aborted, in 1996 SRI started making and selling its own line of regulators. In response, Flotec sued SRI and alleged that SRI had misappropriated its trade secrets. However, since Flotec took no explicit steps to protect the secrecy of its data during and after the negotiations (*i.e.*, warning about confidentiality and demanding their return afterwards), the court ruled in favour of SRI.

4.2 The need for IP slack

An important decision regarding IP contribution is whether IP slack (*i.e.*, extra IP) should be deliberately deployed in R&D alliances. Generally, slack refers to “the pool of resources in an organization that is in excess of the minimum necessary to produce a given level of organizational output” (Nohria and Gulati, 1996, p.1246). Hence, IP slack is the IP committed to an R&D alliance that exceeds the minimum requirement for intended joint R&D. Oftentimes firms choose to contribute related but non-essential technologies to the alliance. Although it could be costly and potentially wasteful to do so, the benefits of IP slack should be noted. Research suggests that slack offers a cushion against unforeseen difficult situations and encourages risk-taking behaviour (Singh, 1986). When partners have additional IP to work with, they may become more experimental in R&D activities.

In 2002 IBM and Sony formed an R&D alliance valued at \$400 million for the development of next-generation consumer electronics devices. Under the agreement, IBM threw wide open its IP gates – essentially providing Sony the full range of IBM process technologies. As a SONY executive commented, “from IBM’s perspective, it has never disclosed so much information before to a partner” (Gain, 2002, p.4). The idea was that, given complete access to IBM process IP, SONY would be able to pick and choose the technologies that best suited its need and held the best promise for success.

Despite the positive effects, too much slack tends to diminish discipline over R&D projects. Accordingly, slack is found to have a U-shaped relationship with innovation: optimal R&D performance is associated with a medium level of slack (Nohria and Gulati, 1996). In light of this finding, whenever possible, firms should try to provide a certain level of IP slack in R&D alliances. Less specific and well-defined R&D purposes tend to be associated with a need for greater IP slack. Many R&D alliances are in fact exploratory in nature and R&D purposes emerge and evolve over time. IP slack can be critical in fostering an environment in which creative experiments are encouraged.

4.3 *Interpartner trust*

In selecting IP inputs for R&D alliances, an important factor is the level of trust between the partners. Trust can be viewed as one's belief that the other party will not intentionally take advantage of the first party (Barney and Hansen, 1994). It is a belief in the partner's goodwill to be fair in the relationship. Trust is a valuable element in interfirm relationships, since it significantly lowers the transaction costs and perceived risks in such deals. Partner firms with a high level of trust are able to proceed faster, commit more and collaborate better. In the long run, partners may develop trust through communication, equity preservation, interpartner adaptation, and risk taking (Das and Teng, 1998). In the short run, however, they have to make IP decisions based on their current level of trust.

In general, a lack of interpartner trust would encourage the use of IP with better ownership protection. That is, patents will be preferred over trade secrets. By comparison, trade secrets are more vulnerable to opportunistic learning, as they require the owner to take explicit steps to secure their secrecy. There is a concern that sharing trade secrets with others may undermine the legal status of the trade secrets. Hence, without sufficient trust, firms fear that their partners may opportunistically appropriate trade secrets. Since commitment of IP connotes risk taking, trust helps alleviate partners' fears over opportunism.

4.4 *Intellectual Property Rights (IPR) protection regimes*

Another factor that influences the selection of IP is IPR protection regimes, which differ significantly across countries, regions and trade organisations (Chaudhry and Walsh, 1995; Mansfield, 2000). For example, some countries do not recognise trade secrets at all. In addition, many nations do not recognise the so-called non-compete agreements (Mandell, 2000), which prohibit employees or firms from using trade secrets from their former employer/partner. Although most countries have patent laws, the strict implementation of such laws cannot be taken for granted in every country. Given the differences, it is important that firms involved in international alliances have sufficient prior knowledge about the IPR protection regime of the other country, especially where joint R&D activities actually take place and where R&D outputs are likely to be commercialised.

Like interpartner trust, the strength of an IPR protection regime has significant implications on the IP contribution decision. In weak IPR protection regimes, patents are preferred over trade secrets, following the same reasoning regarding interpartner trust. While trust is one's confidence in the individual partner, one's confidence in the legal environment can be equally important. Finally, in case both the protection regimes and trust in the partner are weak, the firm should choose to either opt out of the deal or employ maximum control mechanisms, which is the topic of the next section.

5 **IP control mechanisms**

Once firms decide on the IP to be shared in an R&D alliance, they may employ various mechanisms to protect their IP inputs and minimise opportunistic learning (Norman, 2001). In fact, some researchers believe these two decisions can be simultaneous in

nature (Oxley, 1999), which means that if sufficient protection mechanisms are not possible, the initial IP selection decisions then need to be modified. There exist a range of commonly used IP control mechanisms for consideration.

5.1 Equity arrangement

A widely used control mechanism in R&D alliances is equity arrangement. There are two types of equity arrangements: joint ventures and minority equity investment. While the former involves the creation of new entities jointly owned by the partners (*i.e.*, equity creation), the latter is about firms acquiring minority equity positions in partners (*i.e.*, equity transfer). R&D alliances may take the form of R&D-only joint ventures, or include other functions (*e.g.*, production and marketing) to become full-fledged entities. Minority equity investment can be used to consolidate the relationship and financially support joint research. For example, pharmaceutical firms are often interested in R&D projects of biotech firms, which tend to lack financial and regulatory resources to commercialise their R&D outputs. Thus, joint R&D is made possible when pharmaceutical firms take equity positions in biotech firms.

Another purpose of equity arrangements is to align partners' interests and deter opportunistic behaviour, including opportunistic learning (Gulati, 1995). First, since shared equity leads to aligned interests between partners, it is likely that partners will focus more on the success of joint R&D rather than opportunistically obtaining others' IP. Second, equity positions may serve as hostage that deters opportunistic behaviour. With equity arrangements, it will be harder to get away with appropriating others' IP.

5.2 R&D setup

Besides equity, another structural aspect of alliances is the physical setup of collaboration, which can be either joint or separate. A joint setup requires partners to work together under one roof, while in a separate setup partners would carry out their part of R&D responsibility at their own facilities. For instance, Airbus employs a separate setup so that the four partners primarily conduct R&D in their focused area, such as engine or wing design.

Regarding IP protection, the benefit of the separate setup is that, when firms work separately, they do not have easy access to partners' IP, particularly trade secrets. Less IP exposure reduces the risk of opportunistic learning. Nevertheless, the tradeoff is that certain joint R&D demands seamless collaboration between partners. Thus, although a separate setup promotes IP protection, it may compromise IP generation.

When a joint setup is used, firms also have the options of locating the operation in one of the partners' facilities or in a third place (Baughn *et al.*, 1997). Choosing a third place limits IP exposure to partners, as only the relevant technology and knowledge will be present. In contrast, hosting joint R&D in one's own facility runs the risk of IP overexposure. Another remedy for a joint setup is the use of modules to limit IP exposure to the partner. In an alliance with the French company Snecma, General Electric (GE) modularised their jointly produced engine. Since major engine components were made as 'black boxes', Snecma never had IP access to critical parts of the engine (Roehl and Truitt, 1987).

5.3 *Non-disclosure and non-compete agreements*

In addition to structural measures, firms may also use contractual measures to control the outflow of IP and minimise opportunistic learning. A commonly used contract is the non-disclosure agreement, which requires the partner to keep shared IP confidential and to not disclose to third parties (Fraumann and Koletar, 1999). This is an essential step in securing legal protection for trade secrets, which by definition should be kept classified. To make it more restrictive, the partners may also include non-use clauses that prohibit them from using IP shared with each other in future business activities. For example, when Fleetwood Enterprise, a California-based producer of recreational vehicles, forms R&D alliances with suppliers, it would require two-way non-use agreements so that all documented information from each other will be returned upon alliance termination (Schwartz and Harreld, 2001).

In addition, firms can also have no-hire clauses to prohibit partners from hiring the other's own employees for a period. If the firm were still concerned about creating future competitors through the alliance, a non-compete agreement would require that, in the course and aftermath of the alliance, the partner would not compete in the same area for a certain period of time. Such an agreement minimises the possible gains (and risk) of IP misappropriation. In the Sendo-Microsoft dispute, had non-use or non-compete clauses been adopted, Sendo's IP would not have been used by Microsoft to develop competing products.

5.4 *Monitoring and auditing*

The final type of IP control mechanism is behavioural in nature: it requires partners to diligently monitor and audit the status of contributed IP (Norman, 2001). Specific measures may include the appointment of an IP coordinator in the alliance, periodic IP review within the firm, and meetings between partners regarding the use and flow of IP. Such measures also help insure the proper implementation of contractual non-disclosure agreements.

In case problematic IP practice (such as opportunistic learning) is found, firms should negotiate for the following measures:

- compensation for ill-obtained IP
- new clauses controlling the damages of IP leakages (*e.g.*, non-compete agreements)
- removal of personnel responsible for IP misappropriation
- limiting future IP exposure until trust is developed.

Despite the importance of IP auditing, firms often neglect it owing to compliancy. For example, middle- or small-sized companies often step back and let their partners take charge of IP auditing (Kaltenheuser, 1999). Such a lack of due diligence often leads to non-recoverable IP damages.

6 **Classifying R&D outputs: patents or trade secrets?**

As noted, IP is both inputs to and outputs of R&D alliances. Thus, the performance of R&D alliances is mostly measured by IP outputs generated by collaboration. Notwithstanding other possible objectives (such as reputation effects), the overall goal of R&D alliances is to produce R&D outputs that would lead to competitive advantage for

partner firms. To fully take advantage of the R&D outputs (or innovations) of alliances, an important step in IP management is the governance of R&D outputs (*i.e.*, the third step in Figure 1). Effective IP governance is an integral part of R&D alliances and helps maximise their performance.

There are two related issues regarding IP governance: IP classification and IP ownership decisions. Partner firms need to decide how to classify innovations (*e.g.*, as patents or trade secrets), and who owns what R&D outputs. These are new issues for many firms, since the traditional approach is to conduct R&D in-house. Even for firms with other alliance experience (such as joint marketing), joint R&D still poses a unique challenge, since firms usually do not explicitly manage R&D outputs in other alliance types. As more firms start to use alliances to carry out their most important R&D projects, IP governance issues need to be systematically examined.

While this section covers the IP classification issue, the next section deals with the IP ownership issue. Assuming that R&D outputs are not to be disclosed as public knowledge (an option discussed in the next section), IP can be classified as either trade secrets or patents. As noted, trade secrets and patents differ in terms of the legal protection they receive: while trade secrets enjoy the right of not being misappropriated, patents enjoy the right of not being used by others. Despite the difference, firms often do not pay much attention to this important choice, and even in the legal arena, it is not always clear how to advise a client between patent protection and trade secret protection (Di Grazia, 2001). Factors that influence the choice between the two can be related to the external environment, the nature of innovation, and the organisation itself (see Table 1).

Table 1 Classifying R&D outputs: trade secrets versus patents

	<i>Trade secrets</i>	<i>Patents</i>
Environmental factors		
• Weak IPR protection regime		Preferred (as trade secrets tend to be less recognised)
• Rampant industrial espionage		Preferred (to take advantage of patent protection)
• Need for cross-licensing in the industry		Preferred (to take part in cross-licensing)
R&D output factors		
• Low potential value and/or patentability	Preferred (to avoid patenting cost)	
• Easy to patent overlapping technologies	Preferred (to delay patent interference)	
• Easy to reverse-engineer		Preferred (to claim ownership)
Organisational factors		
• Need to distribute R&D outputs		Preferred (to establish industry standards)
• Limited expertise in managing IP		Preferred (to simplify the task)

6.1 *Environmental factors*

The first external factor is the strength of the IPR protection regime, which is often country specific. Strong IPR protection regimes (such as the USA) usually recognise both patents and trade secrets. In contrast, weak IPR protection regimes tend to recognise patents but not trade secrets; and even when both are recognised, they usually offer weak protection for trade secrets. Hence, it will be safer to seek patent protection in such countries.

Some industries (*e.g.*, high-tech industries such as biotech and aerospace) are known to have a problem with rampant espionage. In such industries, it is riskier to rely on protection for trade secrets, since proving trade secret misappropriation is generally more difficult than proving patent infringement. The burden is to prove that the other party did not independently develop the technology and that it was ill gotten from the plaintiff. Accordingly, if possible, innovations from R&D alliances ought to be patented in industries with serious espionage.

In addition, certain industries (such as semiconductor and electronics) are particularly active for licensing and cross-licensing activities (Grindley and Teece, 1997). Since many firms possess related technologies in a given area, licensing and cross-licensing becomes an essential arrangement. To survive and operate effectively in such industries, firms ought to possess as many patents as possible so that they can take part in cross-licensing and take advantage of the close-knit network. In contrast, trade secrets would not work owing to their limited ability for sharing.

6.2 *Research output factors*

Besides external factors, characteristics of the research output also have significant impact on the choice between trade secrets and patents. First, not all innovations are patentable. The success rate of securing patents is only one in every five patent applications (Colson, 2001). Thus, before filing patent applications, partner firms should conduct research to understand the patentability of their innovations and be aware of the latest development in patent laws. For example, an important development in recent years is the enhanced ability to patent business methods, particularly in the e-commerce market. While Priceline.com patented its reverse-auction method and sued Microsoft for infringement, Amazon.com obtained patent protection for its 'one-click-order' method and prevented Barnes and Noble from using the same method (Williams and Bukowitz, 2001). As such, the issue of patentability continues to evolve and deserves close scrutiny.

Second, the potential value of the innovation also determines if a patent application will be worthwhile. The reason is that applying for patents is an expensive option. Not only do legal fees for filing a single patent run to \$10,000–\$15,000, the cost of maintenance and litigation for the life of a patent often reaches \$250,000 (Carter *et al.*, 1999). Apparently, many R&D outputs do not make enough contributions to the firm to warrant the cost of patent protection. It will be unwise to try to patent R&D outputs without regard for their potential value.

Third, certain innovations belong to areas where it is easy to develop and patent related or overlapping technologies. For example, a slight modification may lead to another patent. Since details about patented technologies are publicly available, competitors may use the information to develop alternative and competing patents. Worse, competitors may create a 'picket fence' by patenting add-on features, incremental

developments and surrounding technologies (Colson, 2001). As a result, competitors would own mutually blocking patents that make it impossible for the firm to market any product without infringing on others' patents. Thus, when facing the risk of patent interference, a firm should keep its innovations as trade secrets, at least for a period long enough for it to develop key related and overlapping technologies. These technologies can then be patented at the same time to preempt competitors.

Fourth, technologies that are easy to reverse engineer are not suitable as trade secrets. Reverse engineering allows competitors to develop the same technologies and then possibly file patents for them, a scenario that can be devastating for a firm. For such innovations, filing patents first (or even publishing the technology) would effectively prevent competitors from taking advantage of the weakness of trade secrets.

6.3 Organisational factors

Organisational factors can also be important in choosing between patents and trade secrets. For instance, patents are preferable to trade secrets when partner firms have a strong need to distribute their R&D outputs. In high-tech industries, it is often necessary to share a particular technology with other firms, including competitors, suppliers and customers, in order to establish the technology as the industry standard. Software firms, for example, gave out their patented software (*e.g.*, Explorer, Netscape and Adobe Reader) for free. In the electronics industry, Toshiba and Philips actively licensed their DVD technologies to others in order to win a standard war. Clearly, since trade secrets by definition cannot be effectively shared with a large group, they are not suitable when the need for technology distribution is high.

Another relevant organisational factor is the firm's experience and expertise in managing IP. Since trade secrets are a relatively new form of IP, they tend to be less understood, especially by firms with limited expertise in IP management. Indeed, firms need to continuously guard the secrecy of their technologies in order to maintain the technologies' status as trade secrets. In misappropriation suits the firm needs to prove that:

- The other firm did not independently develop the technology.
- Adequate steps were taken to protect its secrecy.
- The other firm ill got the trade secret from the firm.

Given the complexity, firms less experienced with IP management may wish to simplify their task by staying with the more traditional patent route.

7 IP governance: ownership decisions

While the first aspect of IP governance is classification decisions, the second aspect is related to ownership decisions regarding R&D outputs. Oftentimes ownership decisions are made early in the formation stage of R&D alliances. However, since R&D activities are exploratory in nature, they often lead to unexpected IP results (such as byproducts) whose ownership cannot be fully pre-specified. In such cases partners need to take an adaptive approach in ownership decisions. Thus, from an implementation perspective, IP ownership is part of the last step in IP management.

7.1 Two determining factors

There are four possible ways to allocate R&D outputs: joint, individual, public and third-party ownership. While a large number of issues may influence the ownership decisions, two factors seem to have particular importance:

- 1 the IP's potential as one's core competencies
- 2 the risk of patent interference.

First, certain IP can be developed into core competencies, which are a firm's most valuable assets that distinguish it from the competition. Since core competencies are key to a firm's success, possessing and developing IP with the potential of becoming core competencies is of strategic importance.

Second, as mentioned earlier, patent interference is the risk that other firms may use patents to block and interfere with the otherwise normal use of IP. If a firm's research output significantly overlaps with others' patented technologies, the firm will not have the freedom to fully take advantage of the IP. The reason is that competitors may not license related patents necessary for commercialisation. The following sections discuss how the two factors jointly influence IP ownership decisions (see Figure 2).

Figure 2 IP ownership decisions in R&D alliances

		Potential as one's core competencies	
		High	Low
Risk of patent interference	High	Joint ownership	Public ownership
	Low	Individual ownership	Third-party ownership

7.2 Joint ownership

A joint ownership means the two partner firms share the ownership of their R&D outputs, or both have the guaranteed right to use the innovations through licensing and cross-licensing agreements.¹ Joint ownership is the preferred approach when both determining factors are of high levels. First, when the IP's potential as core competencies is high, the partner firms would like to possess the IP, rather than publish it as public knowledge (public ownership) or sell it to other firms (third-party ownership). Maintaining ownership of the IP is a prerequisite for developing the IP into core competencies.

Second, when the risk of patent interference is high, partner firms tend to be concerned about the possibility that the party owning the IP may not provide due access to it in the future. Thus, individual ownership will be less desirable. In contrast, joint ownership will be preferred as it provides guaranteed access to IP (sometimes through cross-licensing) and reduces the risk of patent interference. Nonetheless, there is one potential complication when two parties jointly own a patent: each co-owner may license the patent to third parties without the consent of the other co-owner (Sobieraj, 2003). To avoid this caveat, an explicit agreement may be developed to rule out unilateral licensing.

7.3 Individual ownership

With an individual ownership, the partner firms will split the R&D outputs, or one partner may end up owning all IP out of the joint R&D. For instance, in sponsored research between biotech firms and pharmaceutical firms, the sponsoring pharmaceutical firms usually get to own all R&D outputs. An individual ownership comes with no comprehensive licensing agreements to provide mutual access to R&D outputs.

An individual ownership is most suitable when the IP's potential as core competencies is high and the risk of patent interference is low. A high potential to be core competencies makes it important to own the IP rather than publicising or selling it. At least one partner will be motivated to obtain ownership. In addition, a low risk of patent interference means that the R&D outputs can be split into pieces that do not overlap enough to block the commercialisation of one of them. Thus, a shared ownership and/or cross-licensing arrangements are not necessary.

7.4 Public ownership

A public ownership is about publishing the innovation so that it becomes public knowledge. This approach is sometimes called a 'defensive disclosure', since publication makes an innovation unpatentable by anyone, including the competitors (Colson, 2001). It is a feasible strategy only when the innovation does not have great potential as partners' core competencies. Otherwise, publicising the innovation would amount to giving away core competencies.

In addition, defensive publishing is plausible when there is a high risk of patent interference, *i.e.*, other firms are interested in patenting surrounding technologies to block the firm's full use of its own innovation. To break this 'picket fence', the firm may choose to publicise information regarding its innovation, including information related to making incremental developments. As a result, the competitors' ability to patent surrounding technologies will be significantly reduced.

7.5 Third-party ownership

The last IP ownership option is to sell the innovation to a third party, sometimes through a public auction. The purpose of this option is to receive quick returns from the innovation, without making additional investment in it. This option allows the partners to unload the technology at a feasible stage of development. Clearly, selling the innovation is a feasible strategy only if its potential as the partners' core competency is low. Otherwise, even if the cost of further development is high, it is still sensible to retain the technology and then seek additional funding for its development.

Another necessary condition for this option is a low risk of patent interference. When the innovation significantly overlaps with the partners' other key technologies, it is possible that the new owner of the innovation may patent it and use it to block the firms' use of other important IP. Thus, when the innovation is co-mingled with other important IP, a third-party ownership should be avoided.

8 Grasp the process of IP management

Given the tendency to mismanage IP in R&D alliances, the importance of a systematic approach cannot be overstated. To this end, this paper presents a three-step framework for IP management, the steps being IP contribution, IP control and IP governance. Because IP management is an ongoing process, being prepared for the uncertainties and exigencies will prove to be invaluable. A few points to minimise unpleasant surprises in the process are summarised below:

- Always make a conscious decision regarding IP contribution in the alliance. First, the firm should take stock of its relevant IP before forming the alliance. Knowing the difference between various types of IP, such as patents and trade secrets, is essential. Second, the IP need for the alliance is only a starting point in deciding what IP to contribute. The decision should also be based on factors such as interfirm trust and IPR protection regimes.
- Be wary about contributing IP slack to the alliance. While IP slack does contribute to a more experimental environment, it is important to note that IP slack also diminishes discipline in R&D. In addition, an overexposure of IP makes the second step (IP control) more difficult and is often the culprit for IP loss in R&D alliances.
- To deter opportunistic learning, a host of IP control mechanisms should be considered and their use should be examined holistically. The reason is that they may substitute for and/or contribute to each other. For example, when an equity arrangement is used, the risk of IP misappropriation will be significantly lower. As such, the need for monitoring and auditing tends to be lower. In addition, since a joint R&D setup comes with greater IP exposure, there will be a heightened need for non-disclosure and non-compete agreements.
- In classifying R&D outputs, the choice between patents and trade secrets deserves much attention, because an adequate understanding of trade secrets has been lacking. A number of factors related to the environment, the innovation and the firms themselves should be considered. What makes this decision tricky is that oftentimes the factors suggest different choices. A careful deliberation regarding the tradeoff will be in order.
- The ownership decisions for R&D outputs are complex and based on options including joint, individual, public and third-party ownership. Attention on two dimensions (*i.e.*, potential as core competencies and risk of patent interference) is critical and simplifies decision making. For example, a public ownership will be feasible only when the potential for core competencies is low and the risk of patent interference is high.

The proposed IP management framework makes several contributions to the literature and thus has implications for future research. First, it focuses on IP management in a specific alliance type – R&D alliances. Instead of treating all alliances as similar, this paper offers a fine-tuned treatment of the alliance type in which IP management is most critical. Such an approach helps reveal the nuance of IP management in alliances. For example, opportunistic learning is shown to be a significant issue for IP contribution and IP control. Future research may build on this approach and investigate how IP management differs in various alliance types.

Second, this paper calls for attention on trade secrets, an IP type that is less understood by many firms. The detailed comparison between trade secrets and patents shows that trade secrets could be the preferred IP type when it comes to contributing and classifying IP. Such systematic examination helps demystify trade secrets and makes them a real alternative to patents in certain alliance situations. Future research may further examine the longitudinal linkage between trade secrets and patents in alliances. For instance, properly managed trade secrets are expected to contribute to firms' innovation capabilities, which in turn lead to more patents in the long run.

Finally, while the existing literature mostly deals with IP protection, this paper offers a comprehensive framework for IP management in alliances. For example, since the performance of R&D alliances depends on generating and utilising IP outputs, this paper helps understand previously understudied IP governance issues (*i.e.*, IP classification and ownership). The implication is that, since IP is both the inputs and outputs of R&D alliances, IP management should go beyond the narrow topic of IP protection. A balance between IP protection and IP generation is proposed, which represents a new approach in managing IP in alliances. As such, the framework significantly expands the scope of IP management in alliances.

Acknowledgement

The author wishes to thank Gurneeta Vasudeva for reviewing an earlier version of this paper.

References

- Barney, J. (1991) 'Firm resources and sustained competitive advantage', *Journal of Management*, Vol. 17, pp.99–120.
- Barney, J.B. and Hansen, M.H. (1994) 'Trustworthiness as a source of competitive advantage', *Strategic Management Journal*, Winter, Vol. 15, pp.175–216.
- Baughn, C.C., Denekamp, J.G., Stevens, J.H. and Osborn, R.N. (1997) 'Protecting intellectual capital in international alliances', *Journal of World Business*, Vol. 32, No. 2, pp.103–117.
- Buckman, R. (2002) 'Sendo suit accuses Microsoft of theft', *Wall Street Journal*, 24 December, p.B5.
- Carayannis, E.G. and Alexander, J. (1999) 'The wealth of knowledge: converting intellectual property to intellectual capital in co-opetitive research and technology management settings', *International Journal of Technology Management*, Vol. 18, Nos. 3–4, pp.326–352.
- Carter, M., Gosling, P. and Poynder, R. (1999) *Hidden Value: How Intellectual Property Know-How Can Make or Break Your Business*, London: Derwent Information.

- Chaudhry, P.E. and Walsh, M.G. (1995) 'Intellectual property rights: changing levels of protection under GATT, NAFTA and the EU', *Columbia Journal of World Business*, Vol. 30, No. 2, pp.80–91.
- Chi, T. and Roehl, T.W. (1997) 'The structuring of interfirm exchanges in business know-how: evidence from international collaborative ventures', *Managerial and Decision Economics*, Vol. 18, pp.279–294.
- Cohen, W.M., Nelson, R.R. and Walsh, J.P. (2000) 'Protecting their intellectual assets: appropriability conditions and why U.S. manufacturing firms patent (or not)', *National Bureau of Economic Research Working Paper Series*, No. 7552.
- Colson, T.J. (2001) 'Leveraging the power and value of intellectual property', *Ivey Business Journal*, Vol. 65, No. 6, pp.29–33.
- Das, T.K. and Teng, B. (1998) 'Between trust and control: developing confidence in partner cooperation in alliances', *Academy of Management Review*, Vol. 23, pp.491–512.
- Di Grazio, J.A. (2001) 'Professional responsibility considerations of advising the corporate client on the patent/trade secret interface', *Intellectual Property and Technology Law Journal*, Vol. 13, No. 5, pp.5–10.
- Doz, Y.L. (1996) 'The evolution of cooperation in strategic alliances: initial conditions or learning processes?', *Strategic Management Journal*, Summer, Vol. 17, pp.55–83.
- Fraumann, E. and Koletar, J. (1999) 'Trade secret safeguards', *Security Management*, March, pp.63–66.
- Gain, B. (2002) 'IBM Micro throws open the IP gates – to offer Sony complete line of process technologies for next-gen devices', *EBN*, 8 April, p.4.
- Grindley, P.C. and Teece, D.J. (1997) 'Managing intellectual capital: licensing and cross-licensing in semiconductors and electronics', *California Management Review*, Vol. 39, No. 2, pp.8–41.
- Gulati, R. (1995) 'Does familiarity breed trust? The implication of repeated ties for contractual choice in alliances', *Academy of Management Journal*, Vol. 38, pp.85–112.
- Gulati, R. (1999) 'Network location and learning: the influence of network resources and firm capabilities on alliance formation', *Strategic Management Journal*, Vol. 20, pp.397–420.
- Halligan, R.M. (2001) 'Do your secrets pass the test?', *Security Management*, March, pp.53–58.
- Hamel, G. (1991) 'Competition for competence and inter-partner learning within international strategic alliances', *Strategic Management Journal*, Summer, Vol. 12, pp.83–103.
- Hemphill, T.A. (2004) 'The strategic management of trade secrets in technology-based firms', *Technology Analysis and Strategic Management*, Vol. 16, No. 4, pp.479–494.
- Inkpen, A.C. (1998) 'Learning and knowledge acquisition through international strategic alliances', *Academy of Management Executive*, Vol. 12, No. 4, pp.69–80.
- Jaffe, A.B. and Lerner, J. (2004) *Innovation and Its Discontents: How Our Broken Patent System is Endangering Innovation and Progress, and What to Do About It*, Princeton: Princeton University Press.
- Jesitus, J. (2000) 'Keeping: manufacturers are vulnerable to proprietary information losses', *Industry Week*, 21 February, pp.9–10.
- Kaltenheuser, S. (1999) 'Look after you leap: do the due diligence', *World Trade*, Vol. 12, No. 6, pp.50–54.
- Larsson, R., Bengtsson, L., Henriksson, K. and Sparks, J. (1998) 'The interorganizational learning dilemma: collective knowledge development in strategic alliances', *Organization Science*, Vol. 9, No. 3, pp.285–305.
- Mandell, M. (2000) 'Guarding vital trade secrets', *World Trade*, February, p.70.
- Mansfield, E. (2000) 'Intellectual property protection, direct investment and technology transfer: Germany, Japan and the USA', *International Journal of Technology Management*, Vol. 19, Nos. 1–2, pp.3–21.

- Nohria, N. and Gulati, R. (1996) 'Is slack good or bad for innovation?', *Academy of Management Journal*, Vol. 39, pp.1245–1264.
- Norman, P. (2001) 'Are your secrets safe? Knowledge protection in strategic alliances', *Business Horizon*, Vol. 44, No. 6, pp.51–60.
- Oxley, J.E. (1999) 'Institutional environment and the mechanisms of governance: the impact of intellectual property protection on the structure of inter-firm alliances', *Journal of Economic Behavior and Organization*, Vol. 38, pp.283–309.
- Park, S.H. and Russo, M.V. (1996) 'When competition eclipses cooperation: an event history analysis of joint venture failure', *Management Science*, Vol. 42, pp.875–890.
- Powell, W.W. (1998) 'Learning from collaboration: knowledge and networks in the biotechnology and pharmaceutical industries', *California Management Review*, Vol. 40, No. 3, pp.228–240.
- Roehl, T.W. and Truitt, J.F. (1987) 'Stormy open marriages are better: evidence from U.S., Japanese and French cooperative ventures in commercial aircraft', *Columbia Journal of World Business*, Vol. 22, No. 2, pp.87–95.
- Schwartz, E. and Harreld, H. (2001) 'A matter of disengagement', *InfoWorld*, 17 September, pp.38–39.
- Singh, J. (1986) 'Performance, slack, and risk taking in organizational decision making', *Academy of Management Journal*, Vol. 29, pp.562–585.
- Sobieraj, J.R. (2003) 'Protecting your IP in joint R&D projects', *Managing Intellectual Property*, 1 July, p.1.
- Williams, R.L. and Bukowitz, W.R. (2001) 'The yin and yang of intellectual capital management: the impact of ownership on realizing value from intellectual capital', *Journal of Intellectual Capital*, Vol. 2, No. 2, pp.96–108.

Note

- 1 For simplicity, I assume there are two partner firms in an R&D alliance, although multi-partner alliances are common and the discussion is equally applicable to such alliances.